

MAGARIA, NigerWMO: **610910**Lat: **12.983N**Lon: **8.933E**Elev: **1322**StdP: **14.01**Time Zone: **1.00 (E01)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	48.7	51.5	17.9	14.3	85.8	20.7	16.3	83.5	15.8	74.2	12.3	75.1	0.4	40	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.7	107.4	69.4	105.8	69.4	104.1	69.3	79.3	91.9	78.7	91.2	78.0	90.5	4.6	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	144.6	84.4	75.6	140.6	83.8	74.9	137.6	83.4	44.0	91.5	43.2	91.4	42.5	90.6	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 12.2	10.9	9.5	DB	43.1	111.3	2.2	1.2	41.5	112.1	40.2	112.8	39.0	113.5	37.4	114.4	(4)
(5)			WB	37.9	82.6	1.3	2.6	36.9	84.5	36.2	86.1	35.4	87.5	34.5	89.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	69.4	75.7	83.6	89.7	91.9	89.0	84.2	81.7	84.3	84.8	77.7	70.8	(6)	
	DBStd	7.90	4.58	5.63	4.48	3.85	3.40	3.89	3.30	2.84	2.84	2.82	3.89	3.94	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	11	9	0	0	0	0	0	0	0	0	0	0	2	(9)	
	CDD50	11650	603	719	1040	1191	1300	1171	1061	983	1028	1078	830	646	(10)	
	CDD65	6186	147	299	575	741	835	721	596	518	578	613	380	183	(11)	
	CDH74	82395	2211	3854	7986	11279	12835	10254	7133	5310	6640	7642	4767	2484	(12)	
(13)	CDH80	48084	965	2101	5073	7605	8597	6226	3495	2240	3377	4503	2783	1119	(13)	
(14)	Wind	WSAvg	3.3	3.3	3.4	3.4	3.3	4.0	4.8	4.1	3.0	2.5	2.4	3.0	(14)	
(15) Precipitation	PrecAvg	21.5	0.0	0.0	0.0	0.1	0.9	2.5	6.7	7.9	3.3	0.4	0.0	0.0	(15)	
	PrecMax	34.6	0.0	0.6	0.7	1.1	4.1	7.0	13.4	15.9	9.4	2.9	0.0	0.0	(16)	
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.0	0.4	1.2	2.0	0.5	0.0	0.0	0.0	(17)	
	PrecStd	6.3	0.0	0.1	0.1	0.2	1.0	1.5	2.7	3.1	1.9	0.7	0.0	0.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.6	102.8	106.4	109.5	109.4	105.7	99.9	95.5	100.9	103.1	101.0	96.9	(19)	
		MCWB	61.9	65.4	66.7	68.4	70.4	74.1	75.1	78.0	73.9	70.0	66.6	63.8	(20)	
	2%	DB	93.2	99.6	104.4	107.6	107.6	103.3	97.0	93.4	98.0	101.5	99.0	93.1	(21)	
		MCWB	61.0	63.8	65.5	68.0	71.0	74.4	76.1	77.7	75.9	69.8	65.3	61.5	(22)	
	5%	DB	89.7	96.5	102.4	106.0	105.9	101.2	94.9	91.7	95.9	99.7	96.8	90.0	(23)	
		MCWB	59.0	62.7	64.5	67.8	71.4	74.9	76.1	77.3	76.9	70.0	64.4	60.0	(24)	
	10%	DB	85.8	92.6	99.7	104.1	104.0	98.9	92.7	89.9	93.7	97.6	94.2	86.6	(25)	
		MCWB	57.7	61.1	63.5	67.1	72.0	75.2	76.2	77.0	77.1	70.7	63.1	58.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	70.0	72.8	76.8	79.2	79.6	79.7	80.0	80.3	78.8	69.6	65.2	(27)	
		MCDB	93.6	96.4	97.9	96.1	95.8	94.6	92.1	91.2	92.5	91.9	95.6	94.0	(28)	
	2%	WB	61.8	65.5	69.1	75.1	77.5	78.3	78.7	79.0	79.2	77.3	66.7	62.7	(29)	
		MCDB	90.2	94.9	96.5	95.2	96.0	93.6	90.9	90.1	91.3	91.4	94.0	89.8	(30)	
	5%	WB	60.1	63.6	66.8	73.6	76.5	77.4	77.8	78.2	78.6	76.3	65.5	61.0	(31)	
		MCDB	87.6	93.7	97.8	94.8	95.5	92.8	89.7	88.7	90.5	90.3	92.6	87.4	(32)	
	10%	WB	58.3	61.8	64.8	72.2	75.6	76.7	77.1	77.4	77.9	75.3	64.2	59.3	(33)	
		MCDB	84.2	91.3	96.2	94.2	94.6	92.0	88.6	87.3	89.5	88.7	90.9	84.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	34.1	32.8	32.8	29.9	24.7	20.2	17.3	16.1	19.9	27.5	34.7	33.6	(35)	
		MCDBR	39.1	36.8	35.9	31.6	27.5	22.8	19.7	18.7	22.6	30.5	36.3	38.1	(36)	
	5% WB	MCWBR	14.9	13.1	12.7	11.1	9.5	5.5	5.5	6.0	6.5	8.6	11.9	14.3	(37)	
		MCWBR	38.4	35.4	33.4	25.1	23.4	20.2	17.9	17.8	20.2	23.1	35.6	36.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.594	0.655	0.703	0.732	0.699	0.674	0.623	0.591	0.596	0.644	0.563	0.559	(40)	
		taud	1.657	1.522	1.445	1.435	1.510	1.602	1.757	1.825	1.797	1.636	1.699	1.720	(41)	
	Ebn at Noon		226	219	213	207	210	212	224	235	234	219	232	231	(42)	
		Edh at Noon	78	92	102	102	93	84	72	68	70	81	74	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		1767	1897	2030	2093	2063	1964	1860	1818	1957	1940	1843	1725	(44)	
	RadStd		72	54	91	69	82	70	61	61	50	62	58	78	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page